Asymmetric Cross-Modal Attention for Multimodal Representation Learning

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1 Introduction

Multimodal learning aims to jointly model information from heterogeneous data sources such as text, images, audio, and sensor signals in order to improve robustness, generalization, and task performance. Many real-world problems naturally exhibit multimodal structure, where each modality captures complementary aspects of the underlying phenomenon. Effectively integrating these diverse sources remains a central challenge, as modalities often differ in scale, noise characteristics, semantic granularity, and predictive relevance.

Existing multimodal fusion approaches typically rely on early fusion, late fusion, or shared self-attention mechanisms. Early fusion methods concatenate raw or lightly processed features, which can obscure modality-specific structure and struggle with heterogeneous representations. Late fusion, while preserving modality independence, often fails to capture fine-grained cross-modal interactions. More recent attention-based methods attempt to address these limitations by learning shared representations through self-attention or symmetric cross-attention; however, these approaches implicitly assume that interactions between modalities are bidirectional and equally informative. In practice, the influence of one modality on another is often asymmetric, where one modality provides contextual grounding while the other contributes task-specific evidence. Symmetric fusion can therefore dilute directional dependencies and limit interpretability.

To address these challenges, we propose an asymmetric cross-modal attention framework that explicitly models directional interactions between modalities. Each modality is first encoded using a modality-specific encoder to obtain latent representations that preserve modality-dependent semantics. During fusion, we employ two separate cross-attention blocks that operate in opposite directions, allowing each modality to attend selectively to the other. This design enables the model to capture how information flows from one modality to another without enforcing symmetry. The resulting cross-attended representations encode complementary, context-aware features that can be flexibly fused and leveraged for downstream prediction tasks. By decoupling modality en-

coding from asymmetric interaction modeling, the proposed framework offers a principled and interpretable approach to multimodal representation learning.

2 Related Work

Multi modal learning

Fusion mechanisms

3 Proposed Approach

Framework Overview. We consider a multimodal learning framework in which each modality is first encoded independently using a modality-specific encoder. Given raw inputs from two modalities, the encoders map them into latent vector representations $\mathbf{a} \in \mathbb{R}^{d_a}$ and $\mathbf{b} \in \mathbb{R}^{d_b}$.

Asymmetric Cross-Modal Fusion. Instead of applying self-attention, the fusion stage employs two asymmetric cross-attention blocks to explicitly model directional interactions between modalities. In the first block, modality $\mathbf{a}$ serves as the query while modality $\mathbf{b}$ provides the keys and values, producing a cross-attended representation

$$\mathbf{a}^{\leftarrow b} = \mathrm{CrossAttn}(\mathbf{Q} = \mathbf{a}, \mathbf{K} = \mathbf{b}, \mathbf{V} = \mathbf{b}),$$

which captures how elements of modality a attend to and are influenced by modality b. In the second block, the roles are reversed, yielding

$$\mathbf{b}^{\leftarrow a} = \mathrm{CrossAttn}(\mathbf{Q} = \mathbf{b}, \mathbf{K} = \mathbf{a}, \mathbf{V} = \mathbf{a}),$$

thereby modeling how modality b selectively attends to modality a.

Cross-Attended Representations and Downstream Usage. The representations $\mathbf{a}^{\leftarrow b}$ and $\mathbf{b}^{\leftarrow a}$ are context-enriched embeddings in which each modality is updated based on the most relevant features of the other modality, as determined by learned attention weights. These representations encode complementary and asymmetric cross-modal dependencies while preserving modality-specific structure. For downstream tasks, they can be fused into a joint multimodal embedding, for example by concatenation,

$$\mathbf{z} = [\mathbf{a}^{\leftarrow b}\,;\, \mathbf{b}^{\leftarrow a}],$$

or by applying pooling or a learned fusion function such as a multilayer perceptron. The resulting fused representation $\mathbf{z}$ is then passed to a task-specific prediction head for objectives such as classification, regression, or generation.

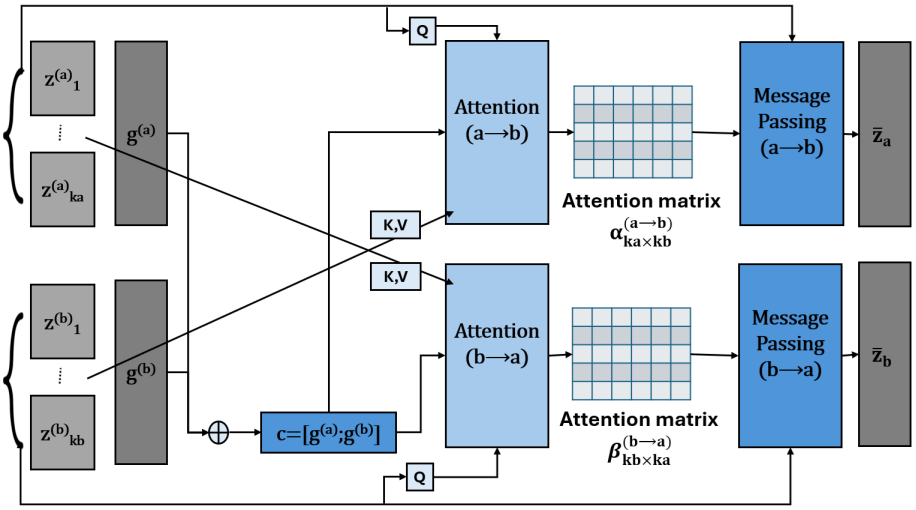

Figure 1: **Fusion architecture:** asymmetric fusion for directional cross-modal information flow.

4 Evaluations

To demonstrate the effectiveness of the proposed approach, we plan to evaluate it on established multimodal benchmark datasets that exhibit heterogeneous modality characteristics and nontrivial cross-modal dependencies. These include vision–language datasets such as VQA or MS-COCO, audio–visual datasets for event recognition, and multimodal tabular–time-series datasets from domains such as healthcare or human activity recognition. A key challenge in these datasets is the imbalance in modality informativeness, temporal or semantic misalignment across modalities, and the presence of spurious correlations when modalities are naively fused. Our framework directly addresses these challenges by learning asymmetric attention patterns that adapt to dataset-specific modality roles. We will quantify effectiveness through standard task metrics, robustness analyses under modality ablation or noise, and comparisons against symmetric attention and fusion baselines to isolate the impact of asymmetric cross-modal modeling.